

Aggregate Functions

Learning Objectives

- Apply COUNT, SUM, AVG, MAX, and MIN functions to calculate data summaries
- Use aliases with aggregate functions for meaningful result columns
- Combine aggregate functions with WHERE clauses for filtered calculations

Engage (5-7 minutes)

A teacher needs to know the class average, highest score, and lowest score. Instead of calculating manually, SQL aggregate functions compute these statistics instantly from the entire table.

Explore (10-12 minutes)

Using a Marks table with StudentID, Subject, and Score columns, write queries to: count total students, find the average score in Maths, find the maximum score in Science, and sum scores for a specific student. Apply aliases to make results readable.

Explain (10-12 minutes)

COUNT() counts all rows, **COUNT(column)** counts non-null values. **SUM** adds numeric values, **AVG** computes the average, **MAX** finds the largest value, and **MIN** finds the smallest. Aliases using **AS** rename output columns. Aggregate functions ignore NULL values except COUNT().

Elaborate (8-10 minutes)

A retail store wants daily sales summaries: total revenue, average order value, number of transactions, and highest single order. Write a query using all five aggregate functions on a Sales table with fields OrderID, Amount, and OrderDate.

Evaluate (5-8 minutes)

Write queries to find: total number of employees, average salary, highest and lowest salary, and total salary expenditure for department "IT". Explain how WHERE filters rows before aggregation.